

Project VSM-1: Speculative Low-Pressure Evaporative Sheet Architecture for Exploration of High-Surface-Area Heat Rejection Regimes

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I. Paradigm & Operational Context

Project VSM-1 is an open-source, directional speculative architecture designed to investigate the physical boundaries of air-side resistance, falling-film hydrodynamics, and boundary layer mechanics in data center heat rejection. It is intended for high-density compute infrastructure (50 MW hyperscale blocks running high-flux AI clusters), treating water chemistry, sheet geometry, and aerodynamics as a single coupled optimization system.

The repository assumes failure is acceptable; the goal is to identify governing constraints rather than validate a production-ready design. It replaces incremental improvements to conventional cooling towers with a research framework for testing whether extreme upstream water purification can unlock ultra-low aerodynamic impedance regimes.

Baseline Comparison Matrix

Cooling Media Geometry

- Conventional: High-density packed PVC or film fills
- VSM-1: Parallel vertical sheet arrays (exploratory 25–40 mm pitch)

Cycles of Concentration (CoC)

- Conventional: 4–8 nominal (up to ~12 with aggressive treatment)
- VSM-1: Elevated theoretical boundary state ($\text{CoC} > 50$)

Aerodynamic Resistance

- Conventional: Pressure drop ≥ 0.50 – 1.0 inches water gauge across packed media
- VSM-1: Target clean-state pressure drop ≤ 0.10 – 0.12 inches water gauge

Water Treatment Paradigm

- Conventional: Continuous biocide/inhibitor dosing with heavy blowdown
- VSM-1: Membrane-based purification with UV and ozone polishing

Primary Parasitic Drivers

- Conventional: High-power fans overcoming dense media resistance
 - VSM-1: Low-head pumps paired with low-RPM, large-diameter electronically commutated fans
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II. Bounded Core Subsystem Engines

1. Water Rectification & Atmospheric Ingestion Boundaries

The system removes mineral scaling (calcium carbonate, silica) as a primary failure mode through aggressive upstream purification.

Treatment Train

Multi-media filtration (10 micron) → hollow-fiber ultrafiltration (0.02 micron) → sodium-cycle softening → reverse osmosis → electrodeionization polishing.

Target inlet conductivity: less than 10–15 microsiemens per cm.

Operating Ceiling

While high purity enables theoretical CoC > 50, real operation is limited by atmospheric contamination:

- airborne particulates
- organic loading
- dissolved gases (CO₂-driven pH shifts)
- material leaching

A secondary reclamation loop (secondary RO or forward osmosis) is required to approach near-zero liquid discharge.

2. Hydrophilic Thin-Film Sheet Vault (“The Sheet Room”)

The system replaces tortuous fill media with vertical sheet arrays to decouple

surface area from aerodynamic drag.

Aerodynamic Target

Clean-state pressure drop: $\leq 0.10\text{--}0.12$ inches water gauge

Air velocity: 2.0–2.5 m/s (horizontal crossflow)

Real systems must also account for support structures, headers, and fouling.

Geometry & Wetting

- Sheet pitch: 25–40 mm
- Surface: corrugated polymer or composite
- Inclination: $35^\circ\text{--}45^\circ$
- Goal: continuous sub-millimeter liquid film

Surface micro-texture is used to induce controlled boundary-layer turbulence without large-scale drag penalties.

3. Hydrostatic Management & Closed Thermal Loop

Water is distributed via gravity-fed weir headers, cascading over sheet arrays into a sloped catch basin (minimum 2% grade) with central drainage.

Low-head centrifugal pumps return fluid through a plate-and-frame heat exchanger, isolating the atmospheric loop from the data center's internal cooling loop (direct-to-chip or immersion systems).

III. Governing Falling-Film & Thermodynamic Physics

1. Falling-Film Hydrodynamics

Film Reynolds number:

$$Re = 4\Gamma / \mu$$

Where:

- Γ = mass flow rate per unit width
- μ = dynamic viscosity

Film thickness (Nusselt approximation):

$$\delta = (3\mu\Gamma / (\rho^2 g \sin\theta))^{(1/3)}$$

Where:

- ρ = fluid density
- g = gravitational acceleration
- θ = sheet angle

Average velocity:

$$u_{avg} = \Gamma / (\rho\delta)$$

Residence time:

$$\tau = H / u_{avg}$$

2. Thermal and Mass Energy Balance

Total heat rejection:

$$Q_{total} = \text{air mass flow rate} \times (\text{air outlet enthalpy} - \text{air inlet enthalpy})$$

Evaporation-driven mass transfer:

$$m_{evap} = K \times A \times (Y_w - Y_a)$$

Where:

- K = mass transfer coefficient
 - A = wetted surface area
 - Y_w = saturation humidity ratio at surface
 - Y_a = bulk air humidity ratio
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IV. Central Existential Problems (Sandbox Boundaries)

1. Surface Tension Frontier: Marangoni Film Collapse

Temperature and concentration gradients create surface tension variations that pull fluid into rivulets, breaking continuous film coverage.

Surface stress relation:

$$\text{surface tension gradient} = (d\sigma/dT \times \text{surface temperature gradient}) + (d\sigma/dc \times$$

concentration gradient)

Failure mode:

- film breakup into rivulets
- reduced surface area
- increased scaling at wet-dry boundaries

Key unknown: minimum wetting rate required to maintain stable film.

2. Biological Surface Colonization

Open, wet, oxygen-rich sheets naturally promote biofilm growth.

Effects:

- increased roughness
 - altered contact angle
 - higher pressure drop
 - reduced heat/mass transfer efficiency
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3. Economic Parasitic Breakeven Problem

High-purity water systems introduce significant energy and capital costs.

Core question:

At what point does purification energy cost exceed gains from reduced air-side resistance?

4. Multi-Phase Fluid Dynamics Drift Model

Key risks:

- droplet entrainment
- aerosolization at weirs
- Weber number instability

Target drift fraction:
< 0.001% of circulating flow

5. Condensate Reclaim Constraints

Water recovery from exhaust air is thermodynamically limited.

Condensation requires a colder surface than dew point, making recovery energy-intensive and climate-dependent. Treated as opportunistic rather than baseline.

V. Degraded-Mode Framework

Highly purified water is chemically unstable and sensitive to contamination.

Failure modes:

- hardness breakthrough (calcium/magnesium scaling)
- rapid pH drift from CO₂ absorption
- corrosion of metals and fittings

Failover behavior:

Switch to low-purity, high-blowdown operation to protect infrastructure.

VI. Falsifiable Validation Hierarchy

Stage 1: Single-Sheet Lab Rig

- measure film thickness evolution
- detect onset of rivulets
- baseline pressure drop

Stage 2: Multi-Sheet Cassette

- crossflow air profiling
- droplet entrainment testing
- fouling simulation

Stage 3: Pilot Vault (50–100 kW)

- full system integration
- CIP validation
- real biofouling data

Stage 4: Industrial Block (500 kW+)

- seasonal performance
 - long-term energy balance
 - full parasitic power accounting
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VII. Regional Validation Roadmaps

Profile A: High-Humidity Temperate (Worst Case)

30°C dry bulb, 25°C wet bulb, 68% RH

- minimal evaporative driving force
- maximum fan and boundary-layer stress

Profile B: Hot Arid (Best Case)

42°C dry bulb, 20°C wet bulb, 13% RH

- maximum evaporative efficiency
- best-case theoretical performance

Profile C: Semi-Arid (Baseline)

35°C dry bulb, 21°C wet bulb, 29% RH

- representative hyperscale deployment conditions
 - economic modeling baseline
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